

Lesson 21C: Matrix Form of System of Two Linear Equations

Criteria	Exceptional (4)	Proficient (3)	Developing (2)	Beginning (1)
1. Augmented Matrix Setup	Correctly translates equations into a 2×3 augmented matrix. Variables are aligned (x,y, =) and coefficients/constants are correctly signed.	Minor sign error or alignment error, but matrix is generally correct and solvable.	Incorrectly identifies coefficients or constants; matrix does not represent the original system.	Unable to form an augmented matrix.
2. Elementary Row Operations	Performs precise row operations (swapping, multiplying, adding/subtracting) to achieve RREF, showing all intermediate steps.	Performs row operations with one or two minor calculation errors that do not fundamentally change the solution type.	Many errors in row operations. Correct steps, but calculation errors lead to incorrect RREF.	Incorrect row operations used (e.g., adding two rows instead of adding a multiple).

3. Identification of RREF	Matrix is in strict RREF: Leading 1s in each row, leading 1 in row 2 is right of row 1, 0s above/below leading 1s.	Matrix is in Row Echelon Form (REF) but not completely reduced (above leading 1s are not all 0s).	Matrix is partially reduced, but pivots are not 1 or not properly positioned.	Matrix is not in RREF or REF.
4. Solution Interpretation	Correctly interprets RREF to state solution: Unique, No Solution, or Infinite Solutions. Correct vector form if needed.	Correct interpretation of the numerical solution, but fails to identify special cases (no/infinite) correctly.	Misinterprets the last column; flips x and y values.	Cannot interpret the RREF matrix.
5. Verification/Checking	Substitutes x and y back into <i>original</i> equations to verify accuracy.	Verifies in one equation but not both, or with minor calculation error.	Attempted to verify but made errors.	No verification performed.